



Loops: Code That Repeats

Facilitation notes, answer key & differentiation

Learning objective

Students write and alter a repeat (count-controlled) loop in a block environment to make a sprite repeat a move/turn action, and describe how changing the repeat count or turn angle changes the figure Bit draws. This is students' first formal work with repeating events in Grade 3.

Ontario curriculum (Math 2020, Strand C3 — Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

This sheet exercises C3.1 (writing a repeat loop in Exercise 1) and C3.2 (altering an existing loop and describing the effect in Exercise 2).

Materials & setup

- A block-coding environment (Scratch-style) projected for the worked example.
- Printed worksheet, one per student. Pencils; optional crayons for the path diagram.
- Optional: a device per pair so students can run Exercise 3 to check their prediction.

How to run the lesson (≈40 min)

1. Hook (5 min): Ask students to clap 4 times. Then say 'repeat 4: clap.' Same result, fewer instructions — that is a loop.
2. Worked example (10 min): Project the repeat-4 square script. Trace it together: count the runs aloud, point to the move and turn each time, watch the square close.
3. Bridge (5 min): Show the Python version side-by-side. Same idea, typed instead of dragged.
4. Guided → independent (15 min): Students do Exercises 1–3. Circulate; ask 'how many times will the inside run?' to surface misconceptions.
5. Consolidate (5 min): Share Exercise 2 answers. Land the big idea: more repeats = more sides; the turn angle decides the shape.

Answer key (all verified by the run-gate)

TEXT

Ex 1 (write): `repeat 3 , turn 120 → triangle (3 sides;  $3 \times 120 = 360$ )`

Ex 2 (alter): `hexagon, 6 sides → 6 repeats make 6 sides; turn 60 closes the shape`

Ex 3 (predict): `120 steps → 3 repeats  $\times$  (20 + 20) = 6 moves  $\times$  20 = 120`

Common errors to watch for

- Counting the blocks once instead of once-per-repeat (Ex 3: forgetting to multiply by 3).
- Confusing repeat count with turn angle — changing repeats changes the number of sides, not the angle.
- In Ex 1, picking a turn that doesn't divide 360 (e.g. turn 100) so the shape never closes.

Differentiation

- Simplify: give Exercise 1 with the turn (120) filled in, so students supply only the repeat count.
- Extend: ask students to write a loop that draws any regular shape — they choose the number of sides and work out $\text{turn} = 360 \div \text{sides}$ (a second loop / a 2nd parameter).

Success indicator

A student meets the goal when they can both WRITE a repeat loop that draws a named shape (Ex 1) and EXPLAIN how changing the repeat count changes the figure (Ex 2) — the C3.1 + C3.2 pairing.